

Question1

Q1.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Question 1</title>
</head>
<body>

  <h1>Question 1 - Student Search System</h1>

  <script src="q1.js"></script>

</body>
</html>
```

Q1.js:

```
// Question 1: Student Search System

let students = [
  "Ali",
  "Sara",
  "Ahmed",
  "Ayesha",
  "Hamza",
  "Sara",
  "Bilal"
];

// Check whether Ayesha is present
console.log("Is Ayesha present?", students.includes("Ayesha"));

// Position of first Sara
console.log("First Sara position:", students.indexOf("Sara"));

// Position of last Sara
console.log("Last Sara position:", students.lastIndexOf("Sara"));

// First student whose name starts with A
let firstAStudent = students.find((student) => student.startsWith("A"));

console.log("First student starting with A:", firstAStudent);

// Position of first student starting with A
```

```

let firstAPosition = students.findIndex((student) => student.startsWith("A"));

console.log("Position of first A student:", firstAPosition);

// Last student satisfying a condition
let lastAStudent = students.findLast((student) => student.startsWith("A"));

console.log("Last student starting with A:", lastAStudent);

// Position of last student satisfying the condition
let lastAPosition = students.findLastIndex((student) => student.startsWith("A"));

console.log("Position of last A student:", lastAPosition);

```

Output:

```

Problems  Output  Debug Console  Terminal  Ports

Is Ayesha present? true
First Sara position: 1
Last Sara position: 5
First student starting with A: Ali
Position of first A student: 0
Last student starting with A: Ayesha
Position of last A student: 3

```

Question2

Q2.html:

```

<!DOCTYPE html>
<html>
<head>
  <title>Question 2</title>
</head>
<body>

  <h1>Question 2 - Product Price Organizer</h1>

  <script src="q2.js"></script>

</body>
</html>

```

Q2.js:

```

// Question 2: Product Price Organizer

let prices = [1200, 450, 3000, 750, 1500, 250];

```

```
// Lowest to highest
let lowToHigh = [...prices];

lowToHigh.sort((a, b) => a - b);

console.log("Lowest to highest:", lowToHigh);

// Highest to lowest
let highToLow = [...prices];

highToLow.sort((a, b) => b - a);

console.log("Highest to lowest:", highToLow);

// Original list
console.log("Original list:", prices);

// Reversed version
let reversedPrices = [...prices];

reversedPrices.reverse();

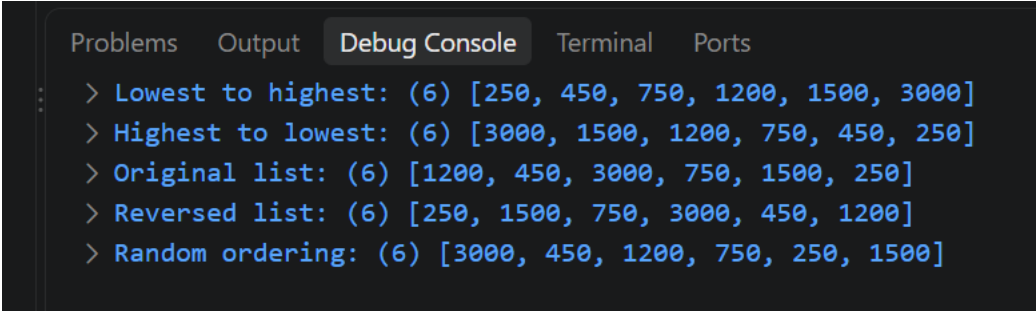
console.log("Reversed list:", reversedPrices);

// Random ordering
let randomPrices = [...prices];

randomPrices.sort(() => Math.random() - 0.5);

console.log("Random ordering:", randomPrices);
```

Output:



```
> Lowest to highest: (6) [250, 450, 750, 1200, 1500, 3000]
> Highest to lowest: (6) [3000, 1500, 1200, 750, 450, 250]
> Original list: (6) [1200, 450, 3000, 750, 1500, 250]
> Reversed list: (6) [250, 1500, 750, 3000, 450, 1200]
> Random ordering: (6) [3000, 450, 1200, 750, 250, 1500]
```

Question 3

Q3.html

```
<!DOCTYPE html>
```

```
<html>
<head>
  <title>Question 1</title>
</head>
<body>

  <h1>Question 3 - Class Result Analysis </h1>

  <script src="q3.js"></script>

</body>
</html>
```

Q3.js:

```
// Question 3: Class Result Analysis

let marks1 = [78, 45, 92, 66, 88, 54, 91, 73];

let marks2 = [81, 69, 95, 58];

// Combine both groups
let allMarks = marks1.concat(marks2);

console.log("Combined marks:", allMarks);

// Create a smaller list
let selectedMarks = allMarks.slice(2, 7);

console.log("Selected marks:", selectedMarks);

// Change one mark in the middle
allMarks.splice(5, 1, 60);

console.log("After changing a mark:", allMarks);

// Total number of marks
console.log("Total number of marks:", allMarks.length);

// Sort from lowest to highest
allMarks.sort((a, b) => a - b);

console.log("Marks from lowest to highest:", allMarks);
```

```
// Reverse the list
allMarks.reverse();

console.log("Reversed marks:", allMarks);

// Arrow function
let showMarks = (marks) => {
    console.log("Final result:", marks);
};

// Call arrow function
showMarks(allMarks);
```

Output:

```
> Combined marks: (12) [78, 45, 92, 66, 88, 54, 91, 73, 81, 69, 95, 58]
> Selected marks: (5) [92, 66, 88, 54, 91]
> After changing a mark: (12) [78, 45, 92, 66, 88, 60, 91, 73, 81, 69, 95, 58]
    Total number of marks: 12
> Marks from lowest to highest: (12) [45, 58, 60, 66, 69, 73, 78, 81, 88, 91, 92, 95]
> Reversed marks: (12) [95, 92, 91, 88, 81, 78, 73, 69, 66, 60, 58, 45]
> Final result: (12) [95, 92, 91, 88, 81, 78, 73, 69, 66, 60, 58, 45]
```

Question4

Q4.html:

```
<!DOCTYPE html>
<html>
<head>
    <title>Question 4</title>
</head>
<body>

    <h1>Question 4 - Employee Record Object</h1>

    <script src="q4.js"></script>

</body>
</html>
```

Q4.js:

```
// Question 4: Employee Record Object

let employee = {
    employeeId: 101,
```

```

    firstName: "Ali",
    lastName: "Ahmed",
    department: "IT",
    designation: "Web Developer",
    salary: 60000
  };

// Dot notation
console.log("First name:", employee.firstName);
console.log("Department:", employee.department);

// Bracket notation
console.log("Designation:", employee["designation"]);
console.log("Salary:", employee["salary"]);

// Add a new property
employee.city = "Hyderabad";

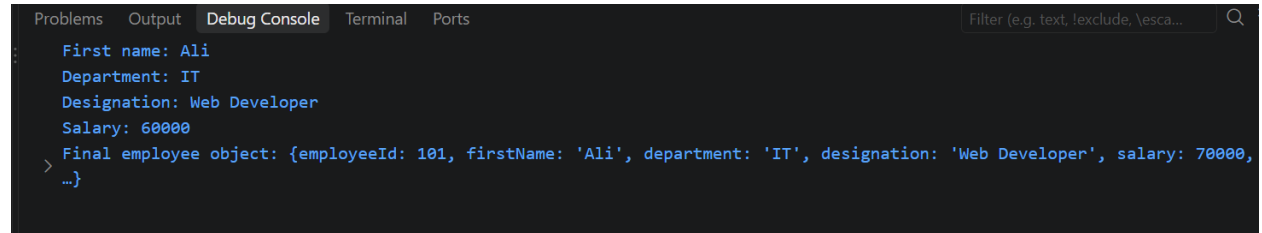
// Change an existing property
employee.salary = 70000;

// Remove a property
delete employee.lastName;

// Display final object
console.log("Final employee object:", employee);

```

Output:



```

Problems Output Debug Console Terminal Ports
Filter (e.g. text, \exclude, \esca...) Q
...
First name: Ali
Department: IT
Designation: Web Developer
Salary: 60000
> Final employee object: {employeeId: 101, firstName: 'Ali', department: 'IT', designation: 'Web Developer', salary: 70000, ...}

```

Question 5

Q5.html:

```

<!DOCTYPE html>
<html>
<head>
  <title>Question 5</title>
</head>

```

```
<body>

  <h1>Question 5 - Employee Object Methods</h1>

  <script src="q5.js"></script>

</body>
</html>
```

Q5.js:

```
// Question 5: Employee Object Methods

let employee1 = {
  employeeId: 101,
  firstName: "Ali",
  lastName: "Ahmed",
  department: "IT",

  // Method 1
  getFullName: function() {
    return this.firstName + " " + this.lastName;
  },

  // Method 2
  getEmployeeInfo: function() {
    return "Employee ID: " + this.employeeId +
      ", Department: " + this.department;
  }
};

// Call methods
console.log("Employee 1 Name:", employee1.getFullName());

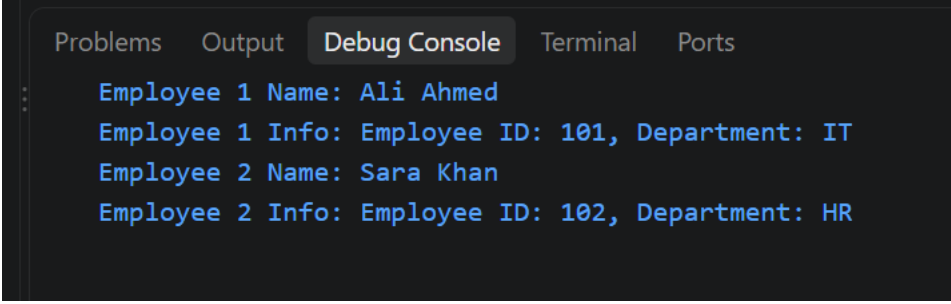
console.log("Employee 1 Info:", employee1.getEmployeeInfo());

// Second employee
let employee2 = {
  employeeId: 102,
  firstName: "Sara",
  lastName: "Khan",
  department: "HR",

  // Method 1
  getFullName: function() {
    return this.firstName + " " + this.lastName;
  }
};
```

```
    },  
  
    // Method 2  
    getEmployeeInfo: function() {  
        return "Employee ID: " + this.employeeId +  
            ", Department: " + this.department;  
    }  
};  
  
// Call methods for second employee  
console.log("Employee 2 Name:", employee2.getFullName());  
  
console.log("Employee 2 Info:", employee2.getEmployeeInfo());
```

Output:



Problems Output Debug Console Terminal Ports

Employee 1 Name: Ali Ahmed
Employee 1 Info: Employee ID: 101, Department: IT
Employee 2 Name: Sara Khan
Employee 2 Info: Employee ID: 102, Department: HR